

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{cases} \partial_t u + u \cdot \nabla u - \Delta u + \nabla P = -\nabla \cdot (\nabla d \odot \nabla d) & \text{in } \Omega \times (0, +\infty), \\ \nabla \cdot u = 0 & \text{in } \Omega \times (0, +\infty), \\ \partial_t d + u \cdot \nabla d = \Delta d + \frac{1}{\epsilon^2}(1 - |d|^2)d & \text{in } \Omega \times (0, +\infty). \end{cases}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} L_{\omega, \mathbf{c}}(V) &= L_{\omega, \mathbf{c}}(V_n) - L_{\omega, \mathbf{c}}(V_n - V) - (L_{\omega, \mathbf{c}}(V_n) - L_{\omega, \mathbf{c}}(V_n - V) - L_{\omega, \mathbf{c}}(V)) \\ &< L_{\omega, \mathbf{c}}(V_n) - 6\mu_{\omega, \mathbf{c}} - (L_{\omega, \mathbf{c}}(V_n) - L_{\omega, \mathbf{c}}(V_n - V) - L_{\omega, \mathbf{c}}(V)) \\ &\rightarrow 0 \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} A_N &= A_1(w_{N-1}) A_{N-1} = \prod_{i=0}^{N-1} A_1(w_{N-i-1}) \\ D_N &= D_1(w_{N-1}) D_{N-1} = \prod_{i=0}^{N-1} D_1(w_{N-i-1}) \\ \begin{aligned}[t] &beginalign*0.3em] C_N &= C_1(w_{N-1}) A_{N-1} + D_1(w_{N-1}) C_{N-1} \\ &= \sum_{i=0}^{N-1} D_i(w_{N-1-i}) C_1(w_{N-1-i}) A_{N-1-i}. \end{aligned} \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathcal{Q}(u) - \mathcal{Q}(\hat{w}) &= \hat{\rho}' \left(U_2 + U_3 \frac{\hat{\zeta}}{|\zeta_0|} + U_\beta \partial_{\hat{\zeta}} H - \lambda w \right) \\ &\geq \hat{\rho}' \left(U_2 + \max\{0, U_3\} + |U_\beta|(1 + \beta)(\zeta_0^2 + (L/d)^2)^{\beta/2} + \lambda \|w\|_{L^\infty(S)} \right). \end{aligned}$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
R_{BCD}^A &= \tilde{R}_{BCD}^A, \\
R_{iBj}^A &= -a_k D_B \partial^A a_l \eta_{ij}^{kl}, \\
R_{AjB}^i &= -a^k D_B \partial_A a_l \eta_{kj}^{li}, \\
R_{jkl}^i &= -a^m \partial^A a_n a_r \partial_A a_s (\eta_{mk}^{ni} \eta_{jl}^{rs} - \eta_{ml}^{ni} \eta_{jk}^{rs}).
\end{aligned}$$